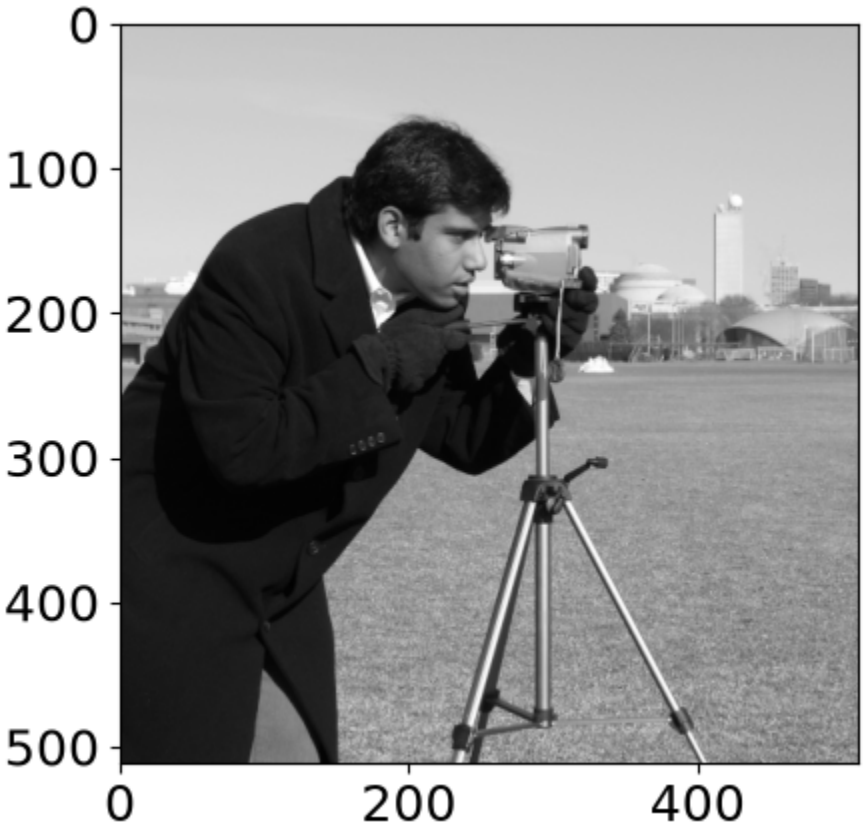
In [5]:
```
import matplotlib.pyplot as plt
import matplotlib
import numpy as np
import skimage as ski
import cv2

from PIL import Image

matplotlib.rcParams['font.size'] = 18
```

In [6]:
```
img = ski.data.camera()

plt.imshow(img, cmap='gray')
plt.show()
```



In [7]:
```
print(type(img))
print(img.shape)
```

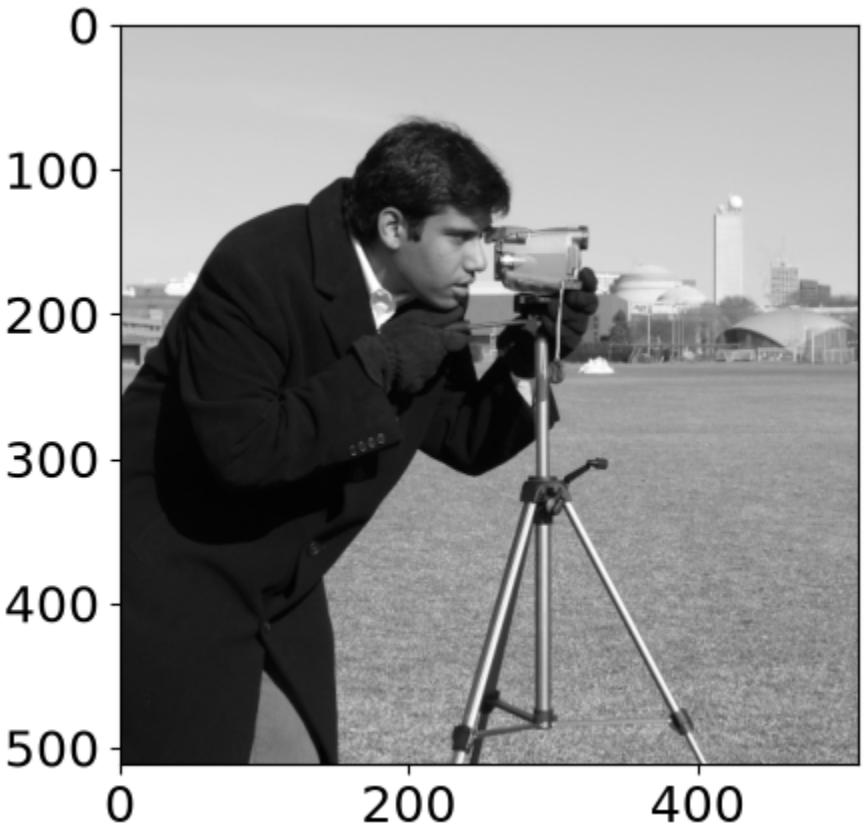
```
<class 'numpy.ndarray'>
(512, 512)
```

In [8]:
```
cv2.imwrite('images/cameraman.tif', img)
```

Out[8]:
```
True
```

In [9]:
```
img_cv = cv2.imread('images/cameraman.tif')

plt.imshow(img_cv, cmap='gray')
plt.show()
```

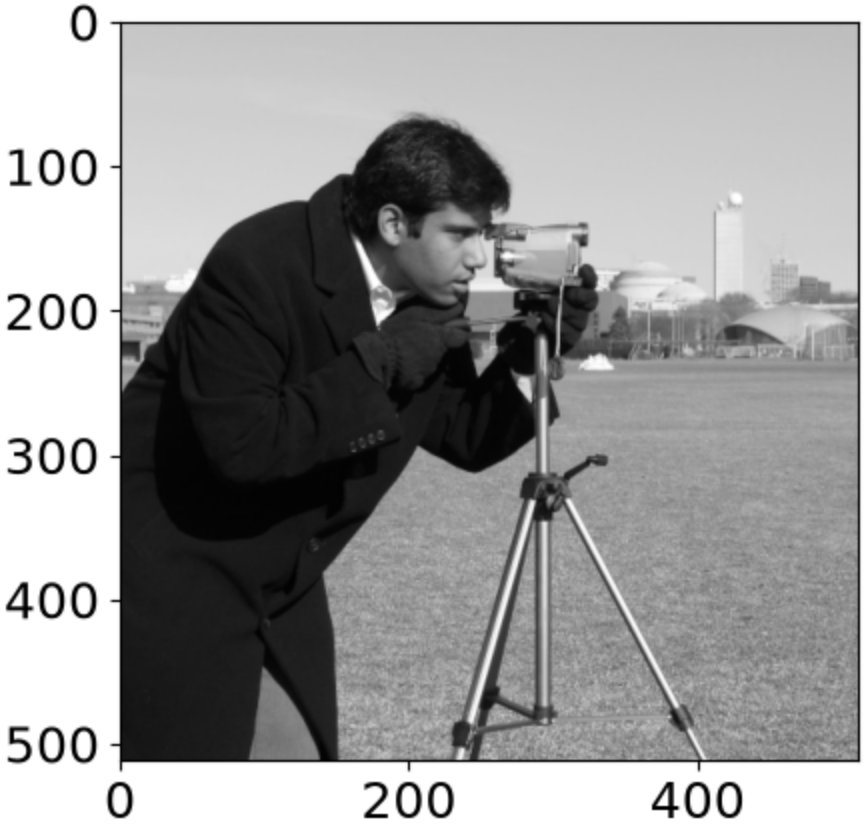


In [10…
```
print(img_cv.shape)
```

```
(512, 512, 3)
```

In [11…
```
img2 = Image.open('images/cameraman.tif')
```

In [12…
```
plt.imshow(img2, cmap='gray')
plt.show()
```



In [14…
```
cv2.imwrite('new_image.jpg', img_cv)
```

Out[14]:
```
True
```

In [15…
```
img2.save('new_image2.jpg')
```

In [16…
```
print(img_cv.shape)
print(img_cv)
```

```
(512, 512, 3)
[[[200 200 200]
  [200 200 200]
  [200 200 200]
  ...
  [189 189 189]
  [190 190 190]
  [190 190 190]]

 [[200 200 200]
  [199 199 199]
  [199 199 199]
  ...
  [190 190 190]
  [190 190 190]
  [190 190 190]]

 [[199 199 199]
  [199 199 199]
  [199 199 199]
  ...
  [190 190 190]
  [190 190 190]
  [190 190 190]]

 ...

 [[ 25  25  25]
  [ 25  25  25]
  [ 27  27  27]
  ...
  [139 139 139]
  [122 122 122]
  [147 147 147]]

 [[ 25  25  25]
  [ 25  25  25]
  [ 26  26  26]
  ...
  [158 158 158]
  [141 141 141]
  [168 168 168]]

 [[ 25  25  25]
  [ 25  25  25]
  [ 27  27  27]
  ...
  [151 151 151]
  [152 152 152]
  [149 149 149]]]
```

In [17…

```
img_array = np.array(img2)

print(img_array.shape)
```

```
print(img_array)
```

```
(512, 512)
[[200 200 200 ... 189 190 190]
 [200 199 199 ... 190 190 190]
 [199 199 199 ... 190 190 190]
 ...
 [ 25  25  27 ... 139 122 147]
 [ 25  25  26 ... 158 141 168]
 [ 25  25  27 ... 151 152 149]]
```

In [ ]: